

Sprint 001: Abstract ThermodynamicStructure Class Hierarchy & Strict Thermodynamic State Interfaces

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Web of Life Simulation Engine

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Abstract

Living systems and complex ecological structures maintain their highly ordered configurations far from thermodynamic equilibrium by continuously dissipating free energy and exporting entropy. This preprint details Sprint 001 of the *Web of Life* simulation engine (<https://github.com/pascalranoroarijaona/WebOfLife>), which establishes the foundational thermodynamic contract: **RFC 001**. We introduce strict TypeScript interface contracts for energetic, entropic, and exergic metrics alongside the abstract base class **ThermodynamicStructure**. This architecture enforces mass-energy conservation (First Law) and unidirectional internal entropy generation ($\dot{S}_{\text{gen}} \geq 0$, Second Law), coupled with exergy destruction tracking via the Gouy-Stodola theorem.

1 Introduction and Systems Ecology Context

Classical ecological simulations frequently model populations and nutrient pools using arbitrary growth curves without explicit mass-energy balance. In contrast, the *Web of Life* engine adopts a strict non-equilibrium thermodynamic perspective. All entities, whether biotic organisms or abiotic pods, operate as open thermodynamic control volumes driven exclusively by external solar irradiance.

Sprint 001 implements the architectural scaffolding required to guarantee that no simulation entity violates physical conservation laws.

2 Governing Thermodynamic Equations

2.1 First Law of Thermodynamics (Energy Balance)

The internal energy U within any control volume (CV) governed by **ThermodynamicStructure** evolves as:

$$\frac{dU}{dt} = \dot{Q}_{\text{in}} - \dot{W}_{\text{out}} + \sum_{\text{in}} \dot{m}_{\text{in}} h_{\text{in}} - \sum_{\text{out}} \dot{m}_{\text{out}} h_{\text{out}} \quad (1)$$

2.2 Second Law of Thermodynamics (Entropy & Exergy)

Entropy evolution accounts for boundary heat/mass exchanges and internal irreversible dissipation ($\dot{S}_{\text{gen}} \geq 0$):

$$\frac{dS}{dt} = \frac{\dot{Q}}{T_{\text{sys}}} + \sum \dot{m}_{\text{in}} s_{\text{in}} - \sum \dot{m}_{\text{out}} s_{\text{out}} + \dot{S}_{\text{gen}} \quad (2)$$

By the **Gouy-Stodola Theorem**, exergy destruction ($\dot{X}_{\text{dest}}$) scales with ambient temperature T_0 and internal entropy generation:

$$\dot{X}_{\text{dest}} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad (3)$$

3 Software Architecture and Type Definitions

The core interfaces defined in `src/thermodynamics/types.ts` establish strict read-only state and flux contracts:

- **ThermodynamicState:** `internalEnergy`, `entropy`, `temperature`, `exergy`, `ambientTemperature`.
- **ThermodynamicFluxes:** `freeEnergyImportRate`, `entropyExportRate`, `internalEntropyGenRate`, `exergyDestructionRate`.

The abstract class `ThermodynamicStructure` (`src/thermodynamics/thermodynamic_structure.ts`) enforces these contracts, providing abstract monad methods for energy import, entropy export, and far-from-equilibrium maintenance, complete with runtime assertions enforcing $\dot{S}_{\text{gen}} \geq 0$.

4 Conclusion

Sprint 001 successfully builds the rigorous thermodynamic core of the *Web of Life* platform. Future sprints will leverage this structure to simulate multi-trophic energy flows and ecosystem resilience.

Repository Reference: <https://github.com/pascalranoroarijaona/WebOfLife>